

Elton Chang

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EDUCATION

- Carnegie Mellon University** May 2026
M.S. in Data Analytics for Science, GPA: 3.97/4.0 Pittsburgh, PA
- Relevant Coursework: Competition Programming, Large Scale Computing, Neural Networks & Deep Learning, Machine Learning in Production, Computational Modeling, Computational Linear Algebra
- University of California, Santa Cruz** June 2025
B.S. in Computer Science, GPA: 3.7/4.0 Santa Cruz, CA
- Relevant Coursework: Data Structures & Algorithms, Computer Systems Design & Assembly Language, Computer Architecture, Applied Machine Learning, Web Applications, Probability Theory

EXPERIENCE

- Undergraduate Research Assistant** September 2024-June 2025
University of California, Santa Cruz Santa Cruz, CA
- Developed scalable and automated data pipelines to clean, standardize, and structured around 600 research files (24 GB) of data, cutting processing time from days of work to under an hour.
 - Optimized a CNN-based image classification models, improving predictive accuracy from 63% to 82%
- Infrastructure Engineer Intern** July 2023-Sep. 2023
Wintec Industries Inc. Newark, CA
- Implemented IaC-defined provisioning for non-prod testing environments, standardizing builds and reducing manual configuration drift across 15+ hosts.
 - Engineered telemetry and alerting across enterprise backup and replication pipelines, turning operational signals into reliability dashboards and cutting anomaly detection time by 40%.
- Software Engineer Intern** July 2022-Sep. 2022
Wintec Industries Inc. Newark, CA
- Built an internal file management system in cloud server for elastic capacity. While building a pipeline allowing automated ingestion, metadata indexing, and lifecycle tracking of company data.
 - Created developer and user-facing documentation for platform architecture, workflows, and system usage, improving onboarding and adoption by 35%.

PROJECTS

- Next Best Action (Collaboration with PNC)**
- An intelligent next-best-action engine for PNC Bank that uses client financial behavior, product holdings, and engagement signals to predict the most relevant high-impact opportunity for advisor outreach.
- Torflix (Open source)**
- A BitTorrent-backed movie library with a FastAPI service that owns swarm I/O, storage, and progressive streaming. With web clients available for catalog, magnet enqueue, and playback while transfers complete.
- Quantifying Meritocracy (Research)**
- Extended the University of Catania's Talent vs. Luck model by introducing a solidarity parameter and applying statistical modeling with agent-based simulations to quantify systemic drivers of inequality.
- Ashe System (Entrepreneurial)**
- A decision-support platform that transforms maintenance, inventory, supplier, certificate, and schedule data into explainable AOG risk scores, operational recommendations, and executive readiness reports.

SKILLS

- Python | TypeScript | Java | C | Go | SQL | React | Next.js | Vue
- Full-Stack Development | RESTful API Design | Database Architecture | System Design